Technical Response to the Third Revision of *Decidability of ALCIRCC5 via Two-Tier Quotient Construction*

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Overall judgment. This version is materially stronger than the earlier drafts. I agree that the move to Union(Desc), the all-phase safety checks around designated and off-chain witnesses, and the explicit chain-unfolding lift lemma are real improvements. However, in my judgment **Theorem 9.1 is still not proved**. Four proof obligations remain open: (i) phasewise safety is not imposed on arbitrary external kernel interfaces; (ii) Step 4 of Theorem 7.1 does not yet derive designated witnesses with the exact demanded relation from the kernel; (iii) the claimed computable size bound is circular; and (iv) the regular-node compression argument still lacks a replacement theorem.

Issue	Status	Comment
Use of Union(Desc) for off-period $\exists PP$-demands	Improved	This fixes a genuine omission from the first two versions and removes one of the earlier witness-coverage objections.
All-phase safety for T4 and V6	Improved	Lemma 4.7 is a worthwhile addition, and it explains why model-extracted stabilized witness relations are compatible with every recurrent phase.
Chain-unfolding lift	Improved	Lemma 8.1 is a real advance: once a finite base network is already valid, constant kernel interfaces can be unfolded without breaking pure RCC5 composition.
Phasewise safety of arbitrary kernel interfaces	Still open	Quotient validity does not require every kernel-to-external edge to be safe against every phase type that will later inherit it.
Exact-relation witness extraction in Theorem 7.1, Step 4	Still open	The proof shows all-phase safety of a stabilized relation S, but T4 requires the stronger fact $\rho(k_\alpha, w) = R$.
Computable bound on quotient size	Still open	The kernel-count estimate depends on $K + L$, while the regular-count estimate depends on K; this does not yet yield a bound in terms of $\lvert C_0 \rvert$.
Regular-node blocking / redirection	Still open	The earlier incorrect domain-identity claim is gone, but no replacement theorem is given that justifies subtree redirection from the weaker blocking key.

1. What the new revision fixes

This third revision is substantially better than the previous ones. In particular, I would now regard the following points as genuine progress.

First, the manuscript no longer treats only the core of a period descriptor as relevant for designated witnesses. The switch from Core(Desc) to Union(Desc) for non-PP kernel demands removes an important earlier gap: phase-specific non-PP demands are at least represented in the quotient data structure.

Second, the all-phase safety requirement for designated witnesses and off-chain PP-witnesses is the right strengthening. The new phase-uniform type-safety lemma explains why, in a genuine model, a stabilized external relation from a recurrent PP-tail to a witness must be safe against every recurrent phase type, not merely against the phase at which the witness was first observed.

Third, the new chain-unfolding lift lemma is a real improvement over the earlier drafts. It isolates a clean RCC5 fact: if one already has a finite composition-consistent network on kernels and regular nodes, then replacing each kernel by an infinite PP-chain while keeping constant external kernel interfaces preserves composition-consistency. That is an important ingredient for any eventual quotient-to-model proof.

Those improvements mean that several of my earlier objections should now be withdrawn. The paper is clearly closer to a proof than the first and second versions were.

2. Remaining obstacles

2.1. Phasewise safety is still missing for arbitrary external kernel interfaces

The crucial new safety conditions appear only in T4 and V6, that is, for designated witnesses and off-chain PP-witnesses. But Step 2 of Theorem 8.3 copies *every* kernel interface uniformly to every point on the unfolded periodic tail:

$$\rho(d_i^{(\alpha)}, e) = \rho(k_\alpha, e).$$

For this to preserve universal restrictions, the validity conditions must ensure that every such relation is safe against every phase type that may appear on that tail.

At present, Definition 6.2 does not require that. A schematic counterexample is immediate. Let

$$\mathrm{Desc}_\alpha = (\tau_A, \tau_B)$$

with both phases containing $\exists PP.\top$ so that the descriptor is admissible, and let only τ_A contain $\forall DR.\neg P$. Take a regular node m with $P \in tp(m)$, and let the quotient assign

$$\rho(k_\alpha, m) = DR.$$

Nothing in V5 excludes this, because V5 checks only RCC5 composition. But after the realization step, every unfolded chain element satisfies

$$\rho(d_i^{(\alpha)}, m) = DR.$$

Hence any position of phase type τ_A violates $\forall DR.\neg P$.

The all-phase checks from T4 and V6 do not repair this, because m is neither a designated witness from T4 nor an off-chain witness from V6. What is still needed is a general interface condition of the form

$$\rho(k_\alpha, e) \in \bigcap_{\tau \in \mathrm{Desc}_\alpha} \mathrm{Safe}(\tau, tp(e))$$

for every external node e, and for kernel-kernel edges an analogous all-phase condition on *both* sides. Lemma 4.7 strongly suggests that model-extracted quotients satisfy such a condition, but the present definition of a *valid quotient* does not require it.

2.2. The exact-relation part of designated witness extraction is still unproved

T4 requires, for every non-PP demand $\exists R.C \in \mathrm{Union}(\mathrm{Desc}_\alpha)$, a designated witness w with

$$C \in tp(w) \qquad \text{and} \qquad \rho(k_\alpha, w) = R.$$

Theorem 7.1, Step 4 begins correctly: at large phase positions one can find witnesses with

$$\rho(d_i, w) = R,$$

and the stabilized kernel relation

$$\rho(k_\alpha, w) = S$$

is all-phase safe by Lemma 4.7.

But all-phase safety of S is not enough. The quotient definition needs the stronger equality $S = R$. The manuscript explicitly acknowledges this and tries to prove it only for $R = DR$. Even there, the argument stops short. It concludes that either a stabilized DR-witness exists, or else the relevant phase positions are witnessed only by "fresh" witnesses whose stabilized relation is *not* DR. The second branch does not produce the T4 witness the definition asks for.

Moreover, no analogous argument is supplied for $R = PO$, even though T4 quantifies over all non-PP demands. So the completeness-to-quotient extraction theorem still does not derive T4 in full generality.

To close this gap, one needs either:

1. a separate exact-relation extraction lemma showing that each phase-level non-PP demand admits a witness whose stabilized kernel relation is the demanded relation itself, or

2. a weaker quotient definition, for example a phase-indexed notion of designated witness, followed by a revised realization theorem.

2.3. The claimed computable size bound is still circular

Theorem 9.1 depends on Theorem 7.1 for a computable bound $f(|C_0|)$ on quotient size. But the proof of Theorem 7.1 still does not derive such a bound.

The problem is visible in Step 2. The number of kernels is estimated by the number of descriptor-interface pairs, with at most D descriptors and at most 4^{K+L} external interfaces per descriptor. In other words, the text gives only a bound of the form

$$K \leq D \cdot 4^{K+L}.$$

Step 5 then bounds the number of regular nodes only in terms of K, via the number of blocking classes. So one obtains something of the form

$$L \leq g(K).$$

This is not yet an upper bound on K and L in terms of $|C_0|$; it is a circular dependency between the unknown quantities that are supposed to be bounded.

Until that circularity is broken, Theorem 9.1 cannot appeal to a finite enumeration of all quotients of size at most $f(|C_0|)$, because no such explicit bound has been extracted.

2.4. The regular-node compression argument still lacks a replacement theorem

This version does improve on the previous one by deleting the old claim that blocked witnesses have identical V6 domains. That claim was too strong. However, the manuscript still needs a theorem of the following form:

> If two off-chain witnesses have the same blocking key, then the later witness and its subtree may be replaced by the earlier one without destroying the existence of a valid finite quotient.

No such theorem is proved.

The blocking key still records only three pieces of information: the local type, the demanded relation to the parent, and the stabilized relations to the kernels. But in complete-graph semantics, later composition and type-safety constraints also depend on relations to earlier *regular* nodes, not only on relations to kernels. Since those relations are not tracked by the blocking key, it is not immediate that a repeated blocking class on a branch really permits redirection of the remaining subtree.

This matters because Step 5 uses exactly that redirection claim to derive finiteness of the regular part. So even after the welcome removal of the earlier incorrect argument, the size proof still lacks a sound compression theorem.

3. Conclusion

My present view is therefore the following.

The manuscript is now a serious and interesting partial result. The additions of Lemma 4.7 and Lemma 8.1, together with the strengthened T4/V6 conditions, improve the paper substantially. Several of the earlier objections are genuinely fixed.

However, I still do not think the manuscript establishes Theorem 9.1. In particular, the current text does not yet show:

1. that arbitrary kernel interfaces are phasewise type-safe,
2. that Step 4 extracts designated witnesses with the exact demanded relation from the kernel,
3. that the quotient size is bounded by an explicit function of $|C_0|$, or
4. that the regular-node blocking key supports the claimed subtree compression.

So my recommendation remains: **acknowledge the progress, but do not yet treat the decidability of $\mathcal{ALCI}_{RCC5}$ as proved from this manuscript alone.**

4. Suggested route to a complete proof

From my perspective, the most promising repair path is now fairly clear.

1. Strengthen quotient validity so that every kernel-to-external interface is checked phasewise, and kernel-to-kernel interfaces are checked against all phase pairs on both sides.

2. Replace Step 4 by an exact-relation extraction theorem, or else weaken T4 to a phase-indexed witness notion and rewrite the realization theorem accordingly.

3. Rebuild the finiteness proof around a genuinely finite interface alphabet or prove a regular-node compression theorem whose hypothesis records enough context to justify subtree redirection.

If those three steps are carried out successfully, the present manuscript would be very close to a complete proof.

References

[1] Claude (Anthropic), prompted by Michael Wessel. *Decidability of ALCIRCC5 via Two-Tier Quotient Construction: Period descriptors, PP-kernel nodes, and the full tractability of RCC5* (third revision, April 2026).

[2] Michael Wessel. *Qualitative Spatial Reasoning with the ALCIRCC Family - First Results and Unanswered Questions*. Technical Report FBI-HH-M-324/03, University of Hamburg, 2002/2003.